

Problem 1: Put six counters in one pile. The move: take one counter from every pile and put all the counters you took together as a new pile. A pile with no counters left is gone. Make the move. Write the pile sizes in the next box, biggest pile first, like 5-1. Keep making moves and writing until the same arrangement appears in two boxes in a row. Write how many moves that took.

6	→		→		→		→	
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 moves: _____

Problem 2: Run these four starts the same way, one on each strip: 1-1-1-1-1; 5-1; 3-3; 2-2-1-1. Write every arrangement. In the table, write the number of moves each start took and the arrangement it stopped at. Which start took the most moves? Which took the fewest? Find one arrangement, other than the one they stop at, that appears in all five runs from Problems 1 and 2, and circle it in every run.

1-1-1-1-1	→		→		→		→	
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5-1	→		→		→		→	
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3-3	→		→		→		→	
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2-2-1-1	→		→		→		→	
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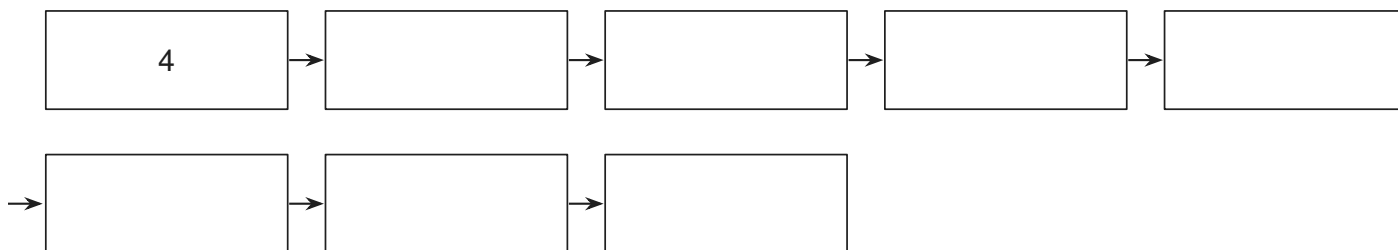
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start	moves	stopped at
6		
1-1-1-1-1-1		
5-1		
3-3		
2-2-1-1		

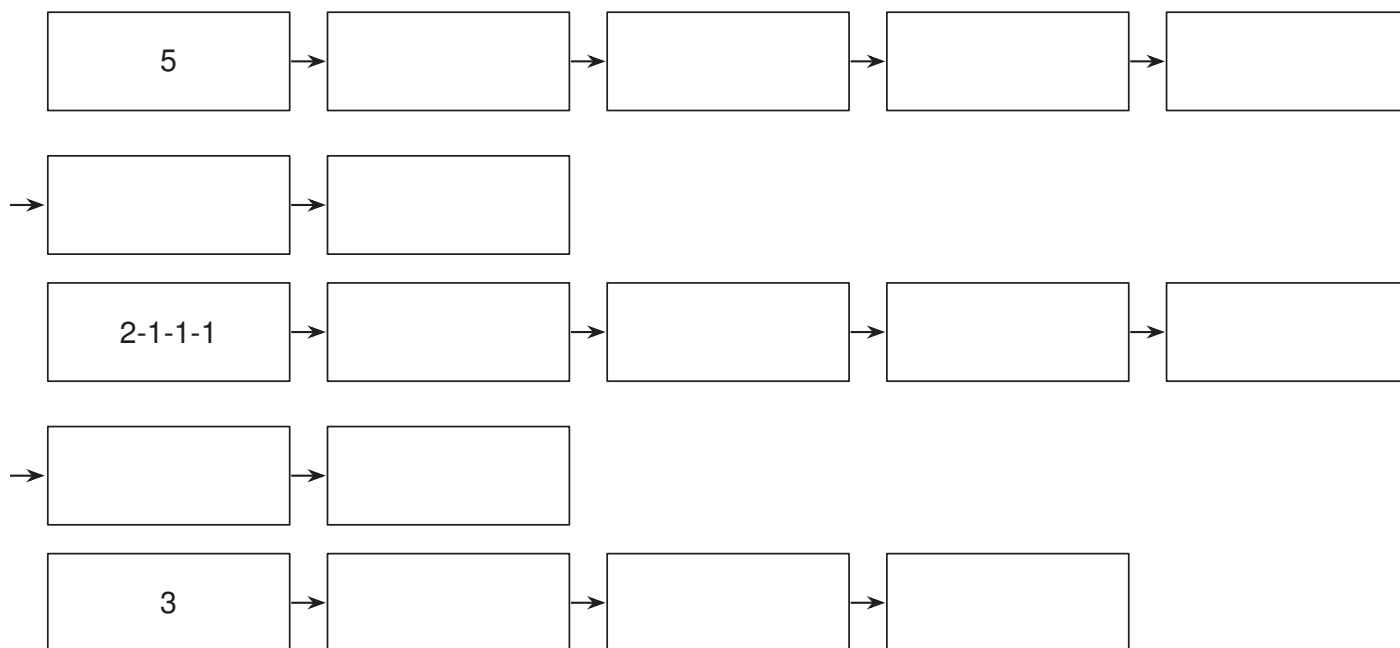
Problem 3: Put four counters in one pile. Make the move seven times and write every arrangement. Circle the first arrangement that appears a second time. How many different arrangements come round again and again? Write the number. Then run 1-1-1-1 for four moves and 2-2 for three moves on the next two strips. Do they get into the same loop?



arrangements in the loop: _____



Problem 4: Put five counters in one pile and run the move until an arrangement appears a second time. Circle the loop. Then run 2-1-1-1 the same way. Does it reach the same loop? Then run three counters in one pile until the same arrangement appears twice in a row. In the table, write R under each number of counters whose runs stop at one arrangement, and L under each number whose runs go round a loop.



3 counters	4 counters	5 counters	6 counters

Problem 5: List every way to split six counters into piles, biggest pile first: 6, then 5-1, and so on. There are eleven. Split the eleven starts between you. Run each start until it stops changing. In the table, write each run and the number of moves it took. Does every start stop at 3-2-1? Which start takes the most moves? Which starts take exactly two moves?

1. _____

2. _____

3. _____

4. _____

5. _____

6. _____
7. _____

8. _____

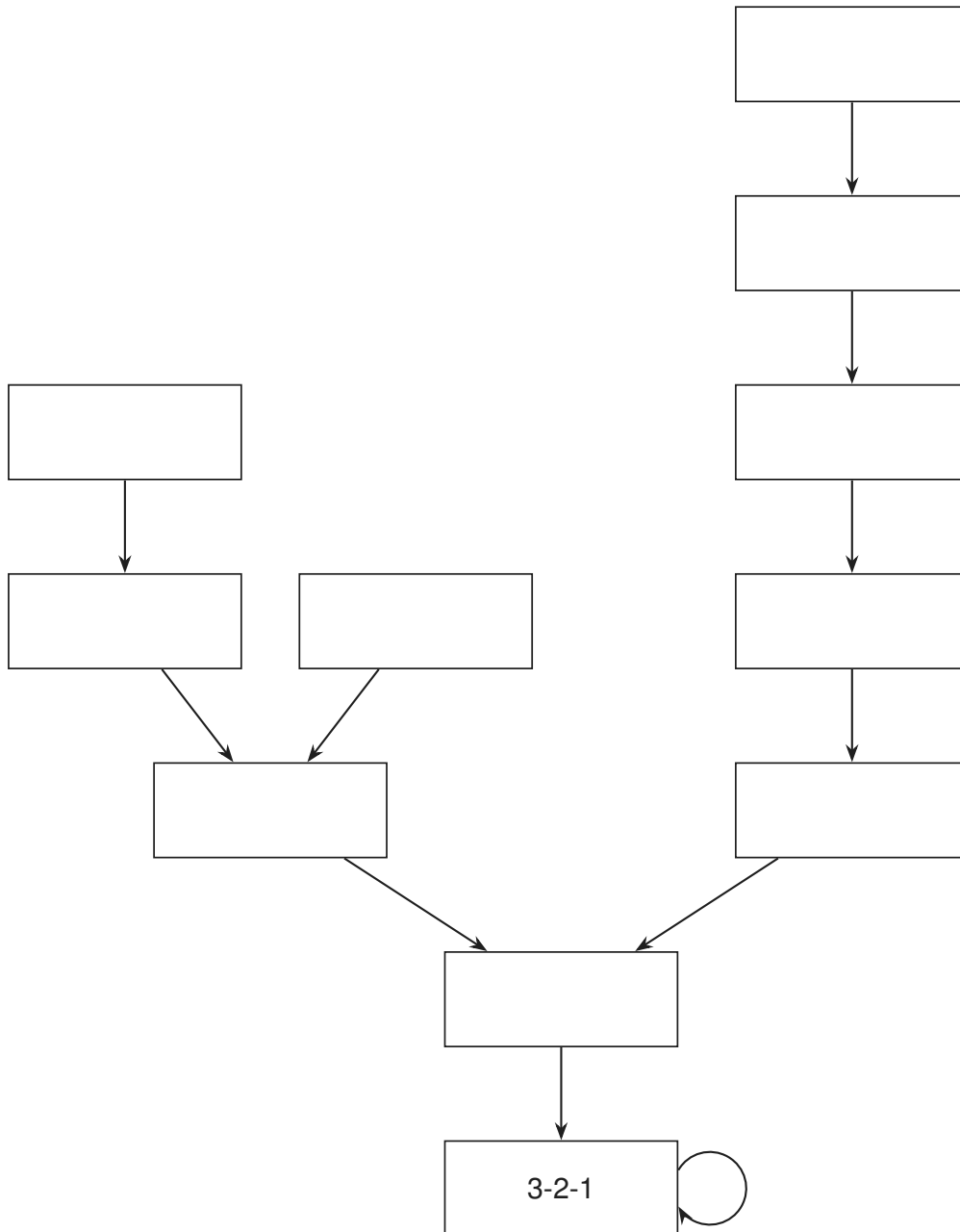
9. _____

10. _____

11. _____

start	run							moves

Problem 6: Write every arrangement of six counters in a box of the map so that each arrow points from an arrangement to what it becomes after one move. Every arrangement goes in exactly one box. Use your table from Problem 5. Then write: which boxes have no arrow coming in; which boxes have two arrows coming in; how many arrows are on the longest path to 3-2-1.



no arrow coming in: _____

two arrows coming in:

arrows on the longest path:

Problem 7: Go back to the runs you wrote. Pick eight different moves. For each one, write the arrangement before the move, its number of piles, and the size of the new pile the move made. What do you notice about the last two columns? Then build 4-4-2 with ten counters. Before each move, write the size the new pile is going to be. Make the move and write the arrangement you got. Do five moves.

arrangement before the move	number of piles	size of the new pile

arrangement	my guess for the new pile	after the move
4-4-2		

Problem 8: Build 3-2-1 and make the move. Write what the pile of 3 became, what the pile of 2 became, what happened to the pile of 1, and how big the new pile is. Now find an arrangement of ten counters that does not change when you make the move. Build it, make the move to check, and write it. Then try to find an arrangement of four counters that does not change. Write every arrangement of four you try and what it became. Write why there is none.

the pile of 3 became: _____

the pile of 2 became: _____

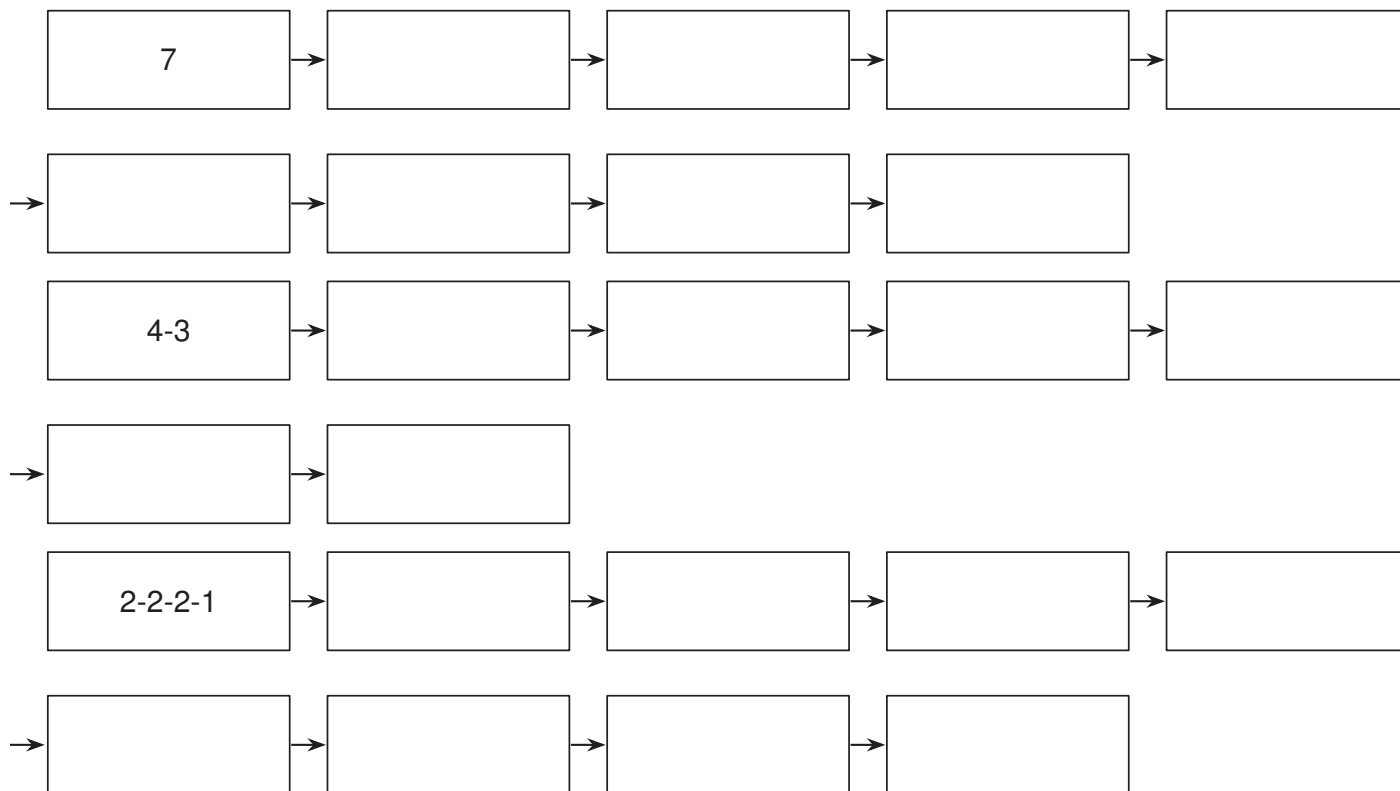
the pile of 1: _____

the new pile has: _____

ten counters that do not change: _____

arrangement of four	after the move

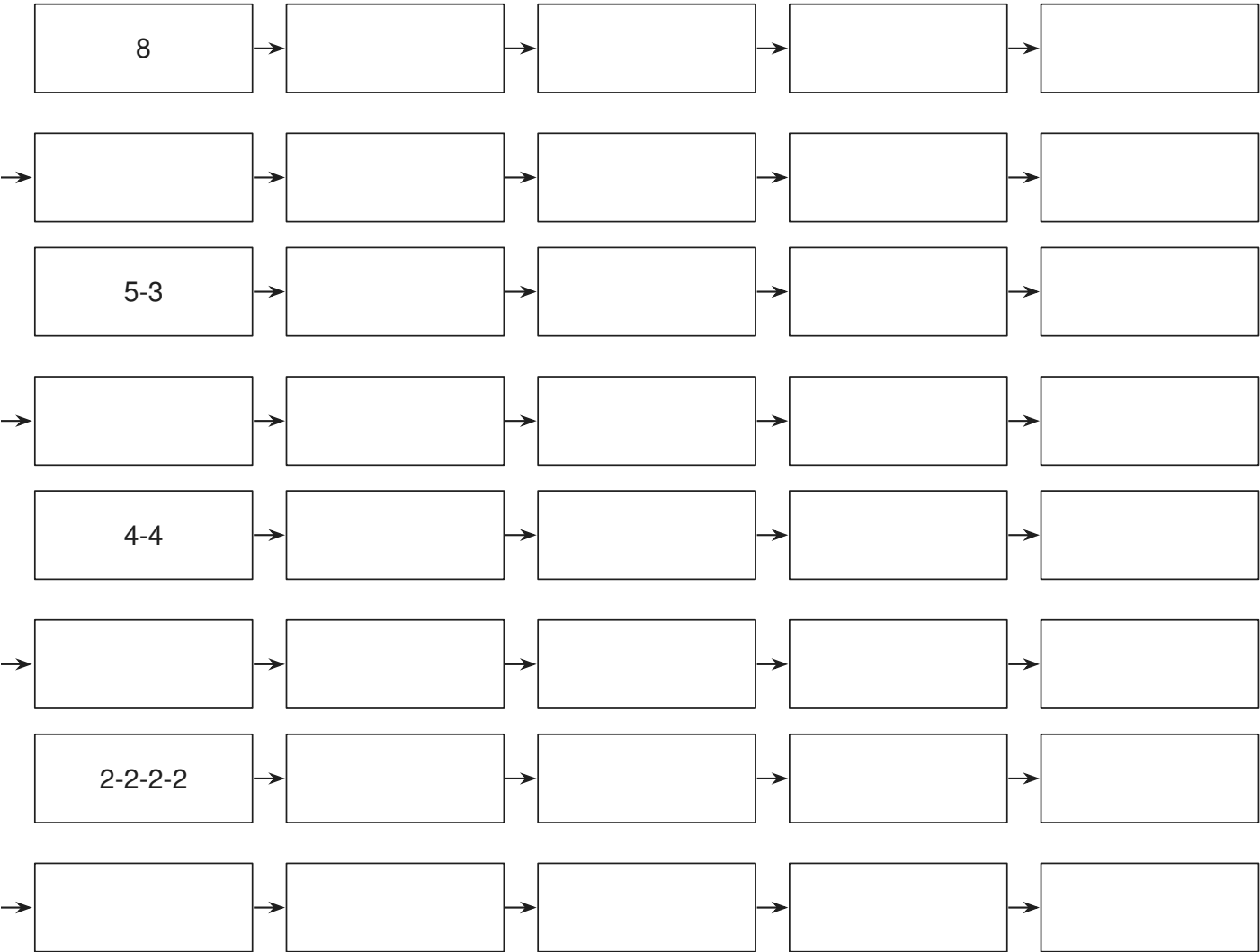
Problem 9: Put seven counters in one pile and run the move until an arrangement appears a second time. Circle the loop. How many arrangements are in it? Then run 4-3 and 2-2-2-1 on the next strips. Do they reach the same loop? Write how many moves each of the three starts takes to get into the loop.



arrangements in the loop: _____

moves to reach the loop, from 7: _____ from 4-3: _____ from 2-2-2-1: _____

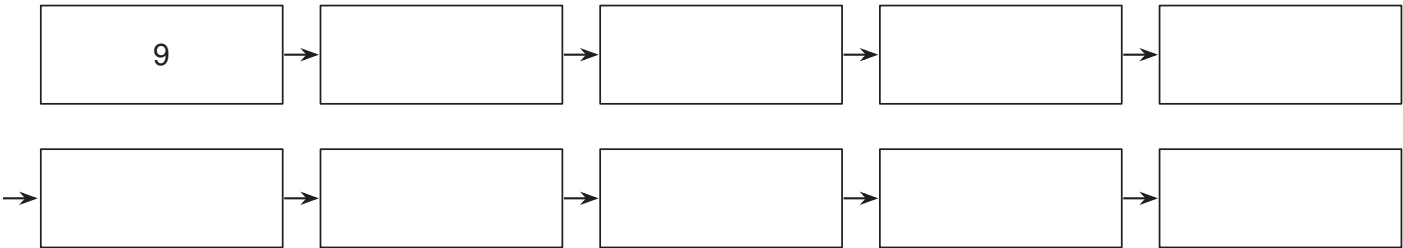
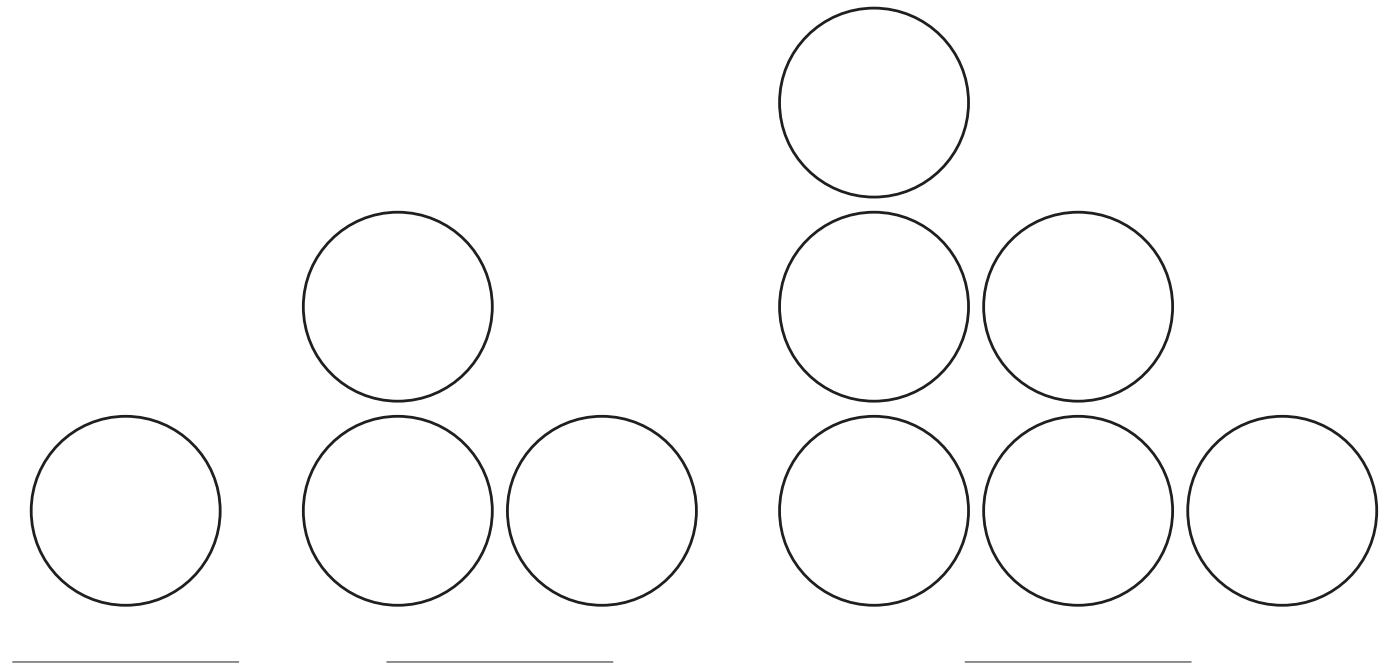
Problem 10: Run eight counters in one pile until an arrangement appears a second time and circle the loop. Run 5-3 the same way and circle its loop. Then run 4-4 and 2-2-2-2. In the table, write for each start the arrangements of the loop it reached and how many there are. How many different loops did you find?



start	arrangements in its loop	how many
8		
5-3		
4-4		
2-2-2-2		

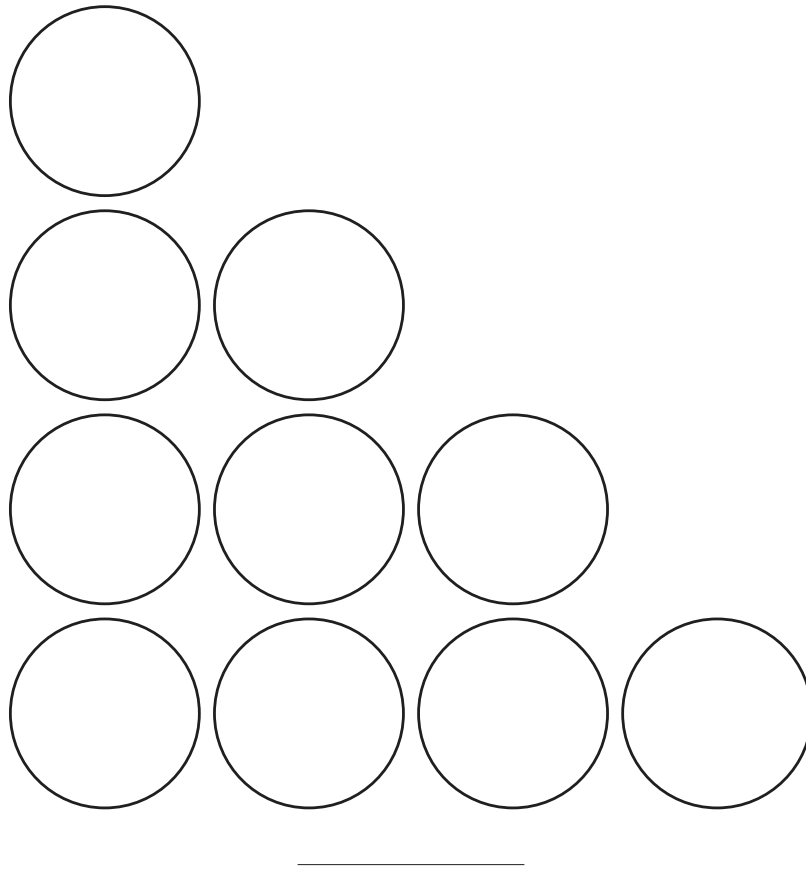
different loops: _____

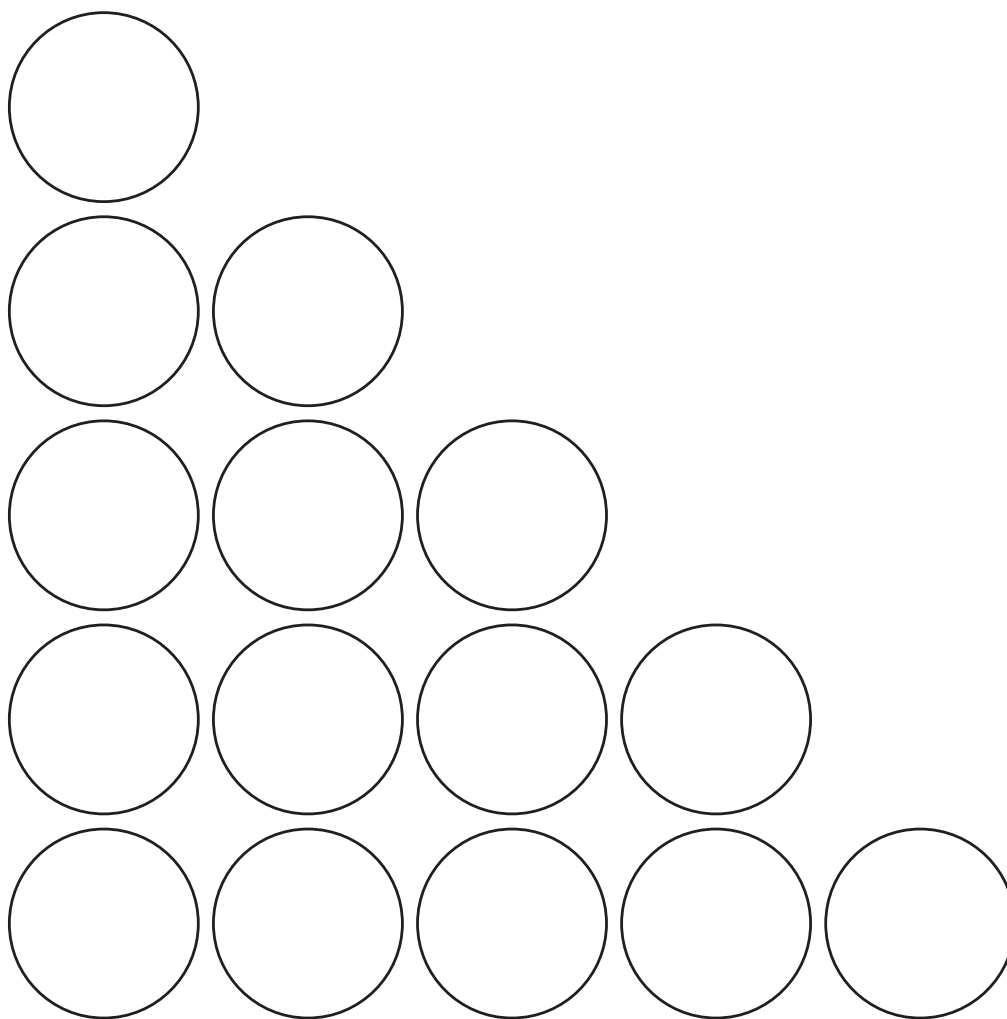
Problem 11: Cover each staircase board with counters and count. Write the five numbers under the boards. Run nine counters in one pile on the strip until an arrangement appears a second time. Then, for each number of counters from 1 to 10, write R in the table if there is an arrangement that does not change under the move, and L if every start goes round a loop. Use your runs, the staircases and Problem 8. For 7, 8 and 9 counters, write your best try for an arrangement that does not change, make one move on it, and write what it became.



1	2	3	4	5	6	7	8	9	10

counters	my try	after one move
7		
8		
9		





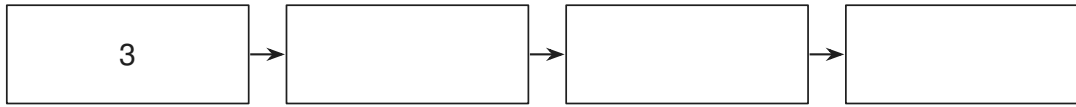
Problem 12: An arrangement that becomes X after one move is called a parent of X . Use the map from Problem 6. Write every parent of 4-2 and every parent of 5-1. Then find every parent of 4-3-2-1 among the arrangements of ten counters, and write them. Then explain, using the size of the new pile, why 2-2-1-1 has no parent.

parents of 4-2: _____

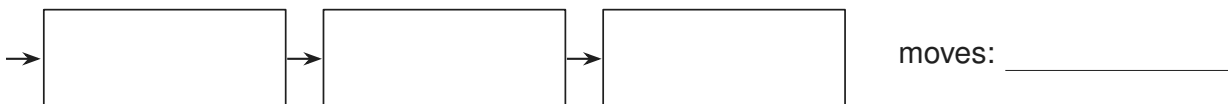
parents of 5-1: _____

parents of 4-3-2-1: _____

Problem 13: Run one pile of three counters until it stops changing and write the number of moves. Do the same for one pile of six, then one pile of ten. Look at the three numbers and write how many moves you think one pile of fifteen will take. Then run one pile of fifteen and write every arrangement. Was your prediction right?



moves: _____



my prediction for fifteen: _____

